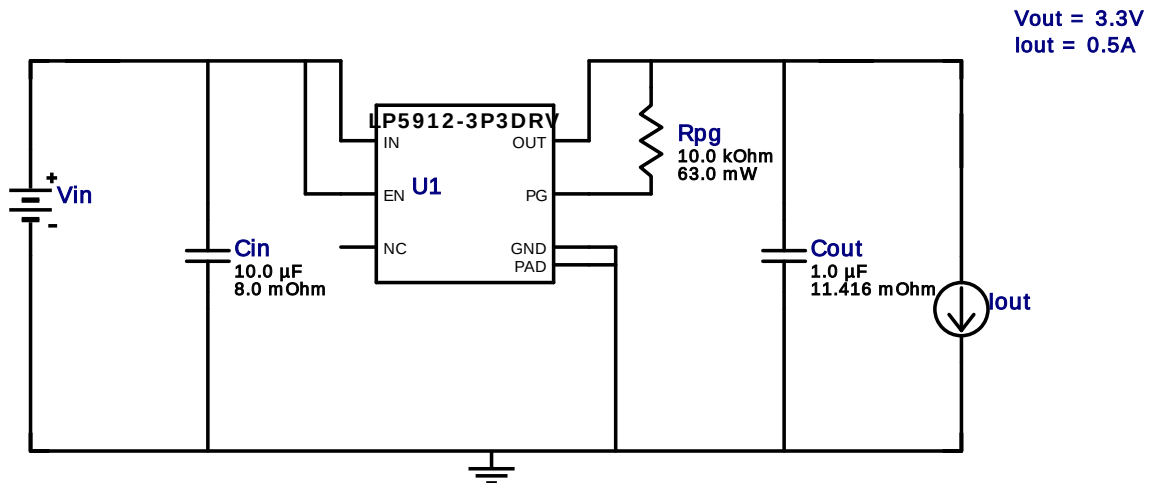


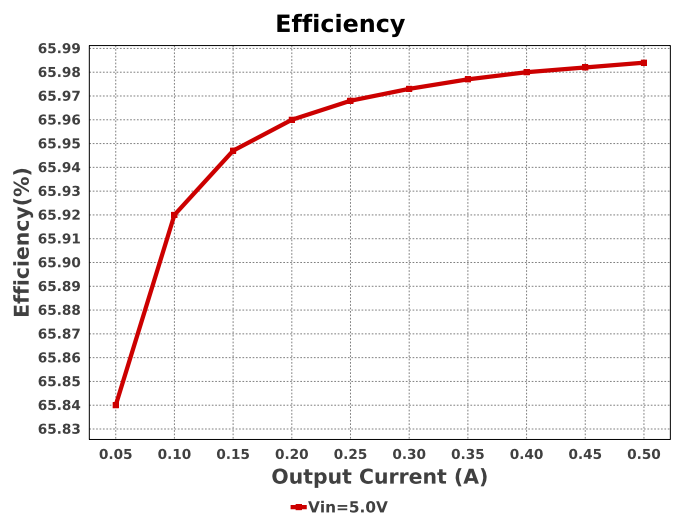
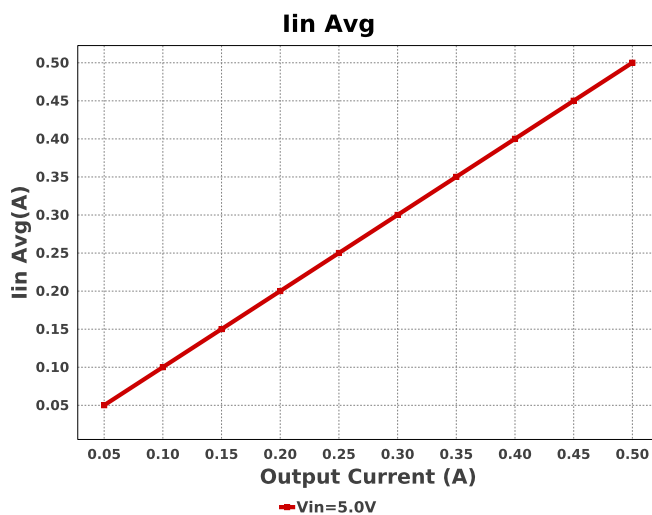
WEBENCH® Design Report

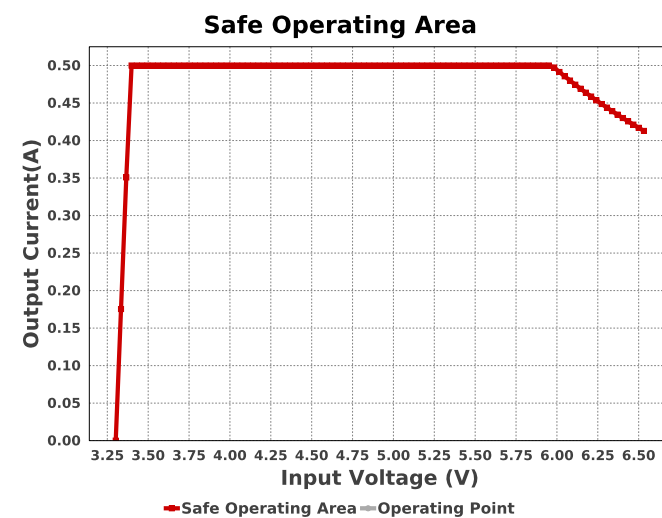
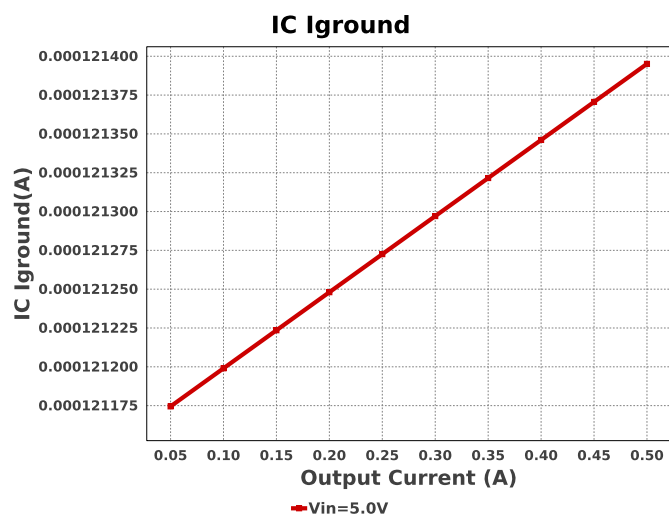
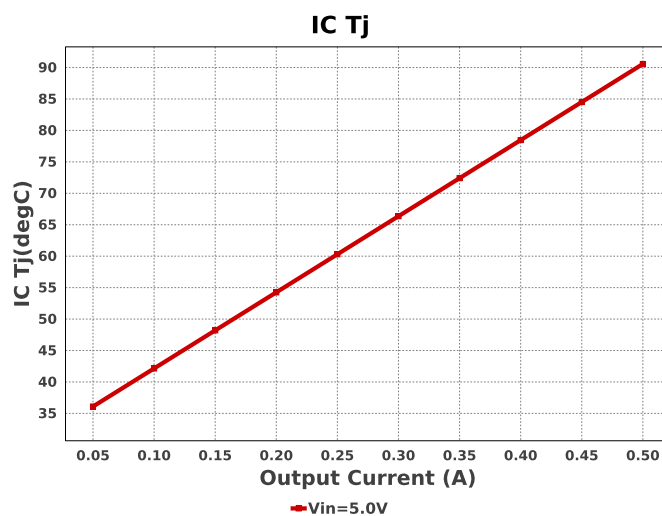
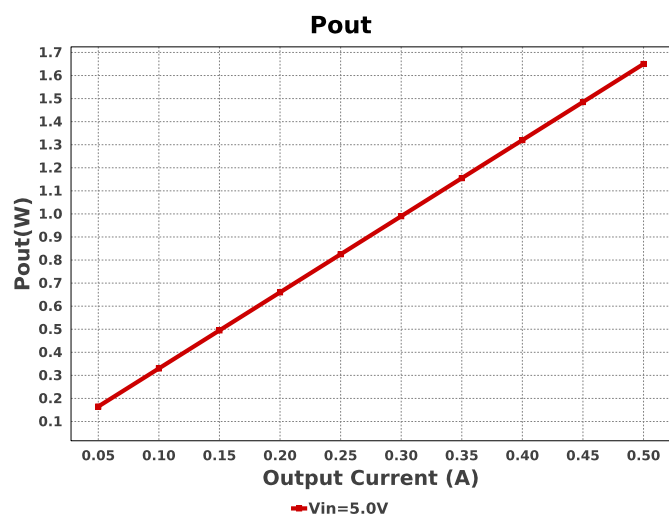
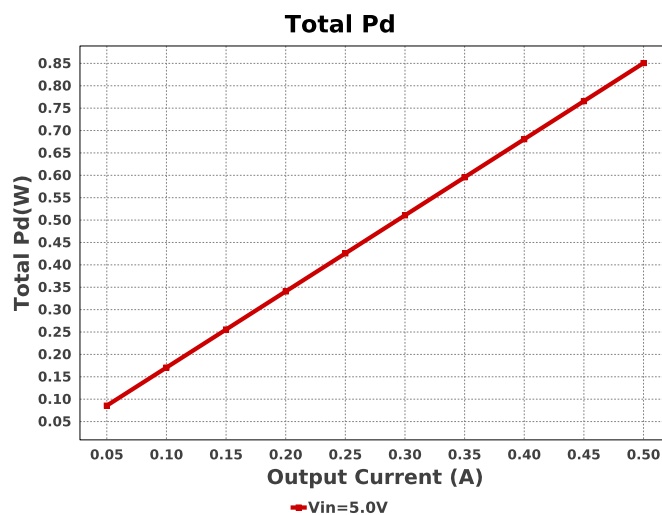
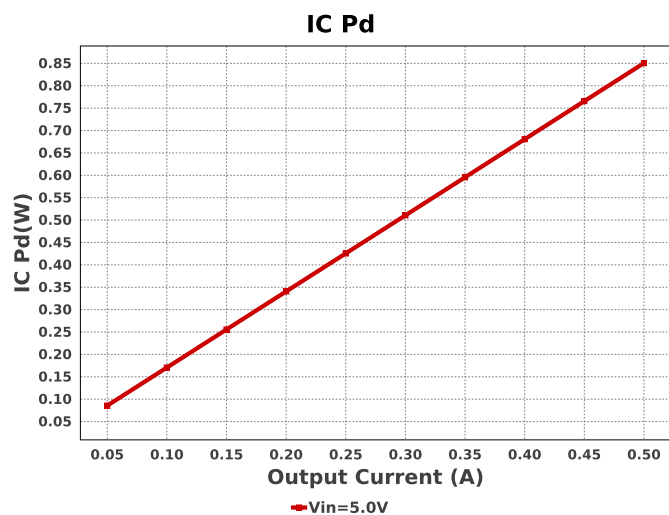
Design : 50 LP5912-3.3DRVR
LP5912-3.3DRVR 5V-5V to 3.3V @ 0.5A

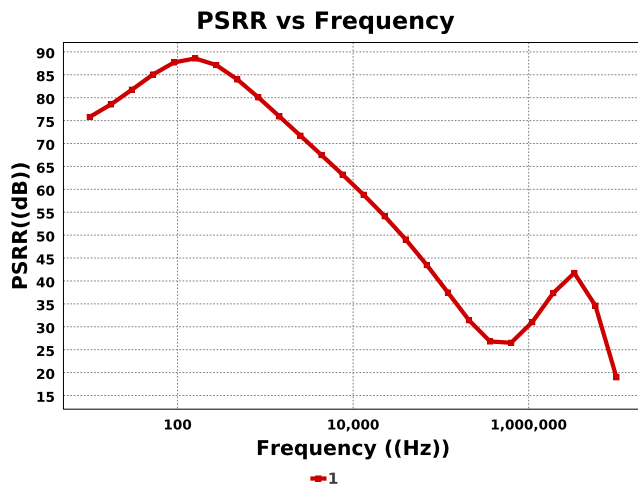


Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cin	MuRata	GRJ155R60J106ME11D Series= X5R	Cap= 10.0 uF ESR= 8.0 mOhm VDC= 6.3 V IRMS= 0.0 A	1	\$0.02	0402_070 3 mm ²
Cout	TDK	C1005X6S1C105K050BC Series= X6S	Cap= 1.0 uF ESR= 11.416 mOhm VDC= 16.0 V IRMS= 1.483 A	1	\$0.02	0402 3 mm ²
Rpg	Vishay-Dale	CRCW040210K0FKED Series= CRCW..e3	Res= 10.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²
U1	Texas Instruments	LP5912-3.3DRVR	Switcher	1	\$0.35	DRV0006A 9 mm ²







Operating Values

#	Name	Value	Category	Description
1.	BOM Count	4		Total Design BOM count
2.	Total BOM	\$0.401		Total BOM Cost
3.	IC Iground	121.395 μ A	IC	IC ground current
4.	IC Pd	850.607 mW	IC	IC power dissipation
5.	IC Tj	90.563 degC	IC	IC junction temperature
6.	IC Tolerance	66.0 mV	IC	IC Feedback Tolerance
7.	ICThetaJA	71.2 degC/W	IC	IC junction-to-ambient thermal resistance
8.	Iin Avg	500.12 mA	IC	Average input current
9.	IOUT_OP	500.0 mA	Op Point	Iout operating point
10.	Input Ripple Frequency	100.0 kHz	Op Point	Input Source Ripple Frequency for PSRR Calculation
11.	PSRR est.	-39.5 dB	Op Point	Power Supply Rejection Ratio estimated
12.	VIN_OP	5.0 V	Op Point	Vin operating point
13.	Total Pd	850.607 mW	Power	Total Power Dissipation
14.	Efficiency	65.984 %	System	Steady state efficiency
15.	FootPrint	18.0 mm ²	System Information	Total Foot Print Area of BOM components
16.	Pout	1.65 W	System Information	Total output power
17.	Vin p-p	50.0 mV	System Information	Input Source ripple voltage
18.	Vout	0.0 V	System Information	Operational Output Voltage
19.	Vout Tolerance	2.0 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
20.	Vout p-p	529.607 μ V	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	5.0	Maximum input voltage
VinMin	5.0	Minimum input voltage
Vout	3.3	Output Voltage
base_pn	LP5912-3.3	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

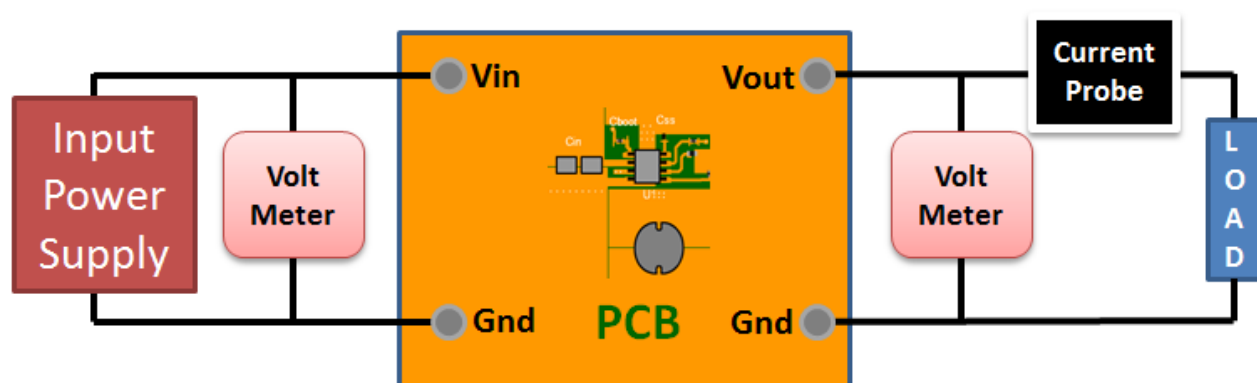
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 5.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

- 1.
2. Master key : DC945615436D3D69B25351734F2F9948[v1]
3. **LP5912-3.3** Product Folder : <http://www.ti.com/product/LP5912> : contains the data sheet and other resources.

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